

Fault-Tolerant Gamma Interconnection Networks

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Abstract

An interconnection network with redundant paths is desirable for high-performance multiprocessor systems, because it can tolerate faults by routing requests through alternative paths, achieving high reliability. The Gamma interconnection network (GIN) is composed of 3×3 basic building blocks, with interconnecting patterns between stages following the plus-minus-2' functions. There is only a unique path from any source (S) to a destination (D) in the GIN when S equals D , and all the multiple paths present between certain (S, D) pairs share a single common route for many stages before starting to fork, causing terminal reliability between such a pair unsatisfactorily low. In this paper, we consider modifications to the Gamma network by altering the interconnecting patterns between stages so as to make totally disjoint paths in existence from any source to any destination, ensuring high terminal reliability between every (S, D) pair. The network is referred to as the fault-tolerant GIN (FTGIN), since it can tolerate an arbitrary single fault. If several building blocks (i.e., 3×3 switches) are fabricated in one chip with VLSI technology, the layout area and pin count are less for the FTGIN than for its GIN counterpart as a result of the change in the interconnecting patterns, giving rise to potential cost reduction. A terminal reliability lower bound of the FTGIN is derived and compared with the terminal reliability of a compatible GIN.

1. Introduction

The multistage interconnection network (MIN) can be employed in a multiprocessor to connect processors and memory modules together, using multiple stages of small crossbar switches of a fixed size. Many MINs make use of 2×2 switches as the basic building blocks, and several of them have been proved topologically equivalent [1]. Such a MIN has one and only one path between any source (S) and any destination (D), and tends to suffer from low reliability, especially for a large sized MIN. As a result, unique-path MINs may be inadequate for high-performance multiprocessors, where reliable communication between every (S, D) pair is often critically essential.

The Gamma interconnection network (GIN) proposed by Parker and Raghavendra [2], like the augmented data manipulator [3] and the inverse augmented data manipulator [4], is one type of MIN composed of 3×3 switches, with interconnecting patterns between stages following the plus-minus-2' functions. While the GIN offers

multiple paths from any source to most of the destinations, there is only a unique path between an (S, D) pair when S equals D , and all the multiple paths existing between certain (S, D) pairs in the GIN share a single common route for many stages before starting to fork, making terminal reliability between such a pair unsatisfactorily low. In Fig. 1, for example, the only two paths between (6, 14) share the same route until reaching stage 3, where the two paths fork, as marked. Therefore, the GIN has many vulnerable communication pairs, regardless of the fact that it has higher hardware complexity than a compatible unique-path MIN, since it needs $N = 2^n$ switches (rather than $N/2$ switches) per stage for connecting N sources to N destinations.

A MIN with redundant paths is desirable for high-performance multiprocessor systems, because it can tolerate faults by routing requests over alternative paths, achieving enhanced reliability. As the GIN is not a fault-tolerant network despite its relatively high hardware complexity, one might wonder if the GIN can be made fault-tolerant without increasing its hardware complexity (or even with its cost reduced, if possible). This work aims to find such profitable designs. It should be noted that one may trivially apply the concept introduced by Adams and Siegel [5] to the GIN to create multiple paths using an additional stage, in a way similar to what was addressed by Lee and Hegazy [6]. However, the complexity of the resulting network is then increased, making this approach practically unfavorable for the GIN whose hardware complexity is already high.

In this paper, we consider modifications to the GIN by altering the interconnecting patterns between stages so that a modified GIN not only provides alternative paths between every (S, D) pair including the case of $S = D$, but also ensures totally disjoint paths from any source to any destination, resulting in high terminal reliability between every communication pair. Such a network is referred to as a fault-tolerant GIN (FTGIN), as it can tolerate an arbitrary single fault. In addition, the FTGIN enjoys potential cost reduction, if several building blocks (i.e., 3×3 switches) are fabricated in one chip with VLSI technology, because the pin count and layout area are less for the FTGIN than for its GIN counterpart.

2. Gamma Interconnection Networks

A size $N = 2^n$ GIN consists of $n+1$ stages, with each stage involving N switches [2]. Every switch in

intermediate stages is a 3×3 crossbar, whereas a switch in the first stage and a switch in the last stage are of sizes 1×3 and 3×1 , respectively. The N sources are numbered from 0 to $N-1$, starting with the top one; so are the N switches in each stage and the N destinations. The stages are numbered from 0 to n rightwards, with 0 for the first (leftmost) stage. Connecting patterns between stages are based on the plus-minus-2' functions, namely, the j^{th} switch in stage i , $0 \leq i < n$, is connected to three switches in stage $i+1$ according to the functions: $f_l(j) = j + 2^i \pmod{N}$, $f_u(j) = j - 2^i \pmod{N}$, and $f_s(j) = j$. The three functions define the lower, the upper, and the straight connections originating from switch j in stage i . A GIN with size 16 is depicted in Fig. 1.

A request from source S in a GIN of size $N = 2^n$ (denoted by GIN_n) can be routed to its destination D under the guidance of an n -digit tag, which represents the difference (modulo N) between D and S . Each tag digit can be 1, 0, or -1, and digit d_i is used at stage i , in a way that the lower (or upper) connection is taken if d_i equals 1 (or -1), and the straight connection is taken when d_i is 0. For the sake of convenience, -1 is denoted by $\bar{1}$. The GIN makes use of the binary fully redundant number system to represent each tag $T = D - S \pmod{N}$. With this number system, a tag $T \neq 0$ has multiple representations, corresponding to multiple paths in the GIN. As an example, 0001, 0011, 0111, and 1111 all represent the same value -1, and they correspond to four possible paths, say, from source 1 to destination 0 illustrated in Fig. 1.

For a given $N = 2^n$, there are 3^n different tag representations, since the tag consists of n digits. The tag values lie in the range of $-(N-1)$ to $(N-1)$, and the values $(i-N)$ and i are equivalent. In Fig. 1, for example, $T = -1 = (15-16)$ has an equivalent tag value of 15, which can be denoted by 1111; the path corresponding to this tag value from source 1 to destination 0 is also shown in the figure. Similarly, for $T = 8$, there are two alternative paths in total, corresponding to the tag representations: 1000 and $\bar{1}000$, as marked in Fig. 1 from source 6 to destination 14.

There are multiple representations for any given tag value using the binary fully redundant number system, except for tag value 0. The total number of different paths (i.e., representations) for a tag value T in GIN_n can be calculated by a recurrence relation [2]:

$$\Phi_n(T) = \begin{cases} \Phi_{n-1}(\frac{T}{2}) & \text{if } T \text{ is even} \\ \Phi_{n-1}(\frac{T-1}{2}) + \Phi_{n-1}(\frac{T+1}{2}) & \text{if } T \text{ is odd.} \end{cases} \quad (1)$$

The number of alternative paths in GIN_4 is listed in Table I, and the number in GIN_6 is given in Table II. It can be seen from the tables that there is only a unique path for tag = 0, and also the number of alternative paths is small for certain tag values. When T equals $N/2$, for example, there are only two alternative paths, and when T is $N/4$ or $3N/4$, there are only three alternative paths. (Notice that this fact holds for any sized GIN.) Moreover, those alternative paths share the same common route

before starting to fork at a later stage. In particular, for the case of $T = N/2$, the two redundant paths share the same route till they reach stage $n-1$, where they fork. (The two paths for T equal to $N/2 = 16/2$ are marked in Fig. 1.) Likewise, if T is $N/4$ or $3N/4$, the three paths follow an identical route until they reach stage $n-2$. As a result, terminal reliability between an (S, D) pair can be unacceptably low when $D - S \pmod{N}$ is $N/2$, $N/4$, or $3N/4$, despite the presence of multiple paths.

All the paths from a source to a destination in the GIN form a ladder structure, based on which the terminal reliability of any tag value can be evaluated [2]. Ladder structures representing all the paths between the (1, 0) pair and between the (1, 4) pair in Fig. 1 are shown in Fig. 5(a). The *terminal reliability* of an (S, D) pair is defined as the probability of having at least one path for communication between the pair. Switches are associated with reliabilities, which denote the probabilities that they are still operational (after a given working period). The failure at a connection is considered to be a switch fault. Faults are assumed to arise randomly and independently.

The terminal reliability results of all tag values in GIN_4 and in GIN_6 are plotted in Fig. 6, where the reliabilities of a 3×3 switch, a 1×3 switch, and a 3×1 switch are assumed to be 0.9, 1.0, and 1.0, respectively. Switches in the first stage and the last stage are assumed fault-free, since such a switch has lower hardware complexity than a regular 3×3 switch and, thus, could be made more reliable. It is desired to have very reliable switches in both end stages, as failures at such a switch disconnect a source or destination from the network, no matter how many alternative paths exist inside the network. As expected, terminal reliability is relatively low if the tag value is equal to 0, $N/2$, $N/4$, $3N/4$, etc. The GIN involves several vulnerable communication pairs, unable to guarantee reliable communication between every source-destination pair. It is thus desirable to ensure the existence of *disjoint* paths between each (S, D) pair, not just multiple paths.

3. Fault-Tolerant GINs

The GIN has larger hardware complexity than a compatible unique-path MIN, as was pointed out in the previous section. In fact, a source can reach any destination in the GIN without using any upper (or any lower) connections. This is reflected in the tag generation process studied by McMillen and Siegel [7], where a tag is called the positive (or negative) dominant tag if its tag bits correspond to only the straight and lower (or upper) connections. Using the third type of connections simultaneously makes it possible to create multiple paths in the GIN. Due to the symmetric connecting style characterized by functions $f_l(j)$ and $f_u(j)$, however, the GIN cannot produce multiple paths if the tag value is 0 and also fails to offer disjoint paths between many source-destination pairs. It appears advantageous by changing the connecting patterns between certain stages such that the upper and the lower connections are no longer symmetric throughout the entire network, if the resultant network guarantees the existence of disjoint paths between every (S, D) pair.

For GIN_n , if the function specified the upper connection from a switch in stage $n-1$ to a switch in the last stage changes to $\hat{f}_{n-1}^m(j) = j - 2^i \pmod{N}$, where $0 \leq i < n-1$, with the remaining functions left unchanged, then multiple paths are created even for the case of $T = 0$. In Fig. 2, for example, $\hat{f}_s^m(j)$ is equal to $(j - 2^1) \pmod{N}$ (the same as $f_s^m(j)$ for stage 1), and two paths exist between source 9 and destination 9, as marked. An additional possible benefit associated with the network given in Fig. 2 is that the wrap-around connections between the last two stages are reduced from 16 (in the GIN illustrated in Fig. 1) to 10. This reduction could translate to a cost savings if several switches are fabricated in one chip using VLSI technology, because fewer pins and less layout area for connections are then needed. As an instance, if four rows of switches in Fig. 2 (inside a dotted rectangle) are housed in a single chip, the pin count for total connections per chip is reduced by 4 (since the pins for upper connections which originate from switches in the second last stage within a dotted rectangle change from 4 to 2, as do the pins for lower connections which terminate at switches in the last stage within a dotted rectangle), and less area is taken by connections between the last two stages. While multiple paths are provided for every (S, D) pair in Fig. 2, there is still no guarantee of disjoint paths from any source to any destination.

When the function $\hat{f}_{n-1}^m(j)$ is chosen to be $(j-1) \pmod{N}$ (the same as $f_0^m(j)$), as shown in Fig. 3(a), two disjoint paths exist from any S to a $D = S$. The two disjoint paths between source 9 and destination 9 in Fig. 3(a) are highlighted. In fact, it will be proved later that disjoint paths are always present from any source to any destination in such a network. Also, choosing $\hat{f}_{n-1}^m(j)$ this way further reduces the pin count and layout area. In the following, we are interested in the $\hat{f}^m$ function of only this kind.

The GIN can generally be modified in a way that one or several consecutive stages immediately before the last stage have the connecting patterns altered. A fault-tolerant GIN of interest is a "modified" GIN such that $\hat{f}_i^m(j)$ changes from $j - 2^i \pmod{N}$ to $j - 1 \pmod{N}$, for all $\alpha \leq i < n$, and the rest of the connection functions remain identical to those in the GIN, denoted by FTGIN_n^α . FTGIN_n^α and FTGIN_n^β are depicted in Fig. 3. An FTGIN has multiple paths between any source-destination pair. All possible paths from source 9 to destination 9 in the two FTGINs given in Fig. 3 are indicated by bold lines.

Each request in an FTGIN_n^α , like that in GIN_n , carries a routing tag of n digits, with each digit equal to $\bar{1}$, 0, or 1. The tag value, again, is the difference between destination and source. The weight of every tag digit is determined by the connecting patterns. Specifically, the i^{th} tag digit, d_i , has the weight of -2^i if d_i is $\bar{1}$ and $i < \alpha$, and of -1 if d_i is $\bar{1}$ and $i \geq \alpha$; it has the weight of 2^i when d_i equals 1. In other words, the tag is represented by means of a "modified" binary redundant number system. Digit d_i is used at stage i when routing the request through an FTGIN: it takes the lower (or upper) connection for $d_i = 1$ (or -1), and takes the straight connection for $d_i = 0$.

Routing complexity in an FTGIN is exactly the same as that in its GIN counterpart.

Calculating Alternative Paths

For a given tag, different "modified" binary redundant number systems yield different amounts of possible representations. Since each representation corresponds to one possible path, the FTGINs under consideration offer varying numbers of alternative paths for a tag. It is interesting to find out the number of redundant paths in FTGIN_n^α , $0 < \alpha < n$. A recurrence equation for the number of redundant paths can be obtained by considering (i) every path which can reach destination D from a switch, say switch SW_c , in stage α (SW_c is called a *candidate switch*), and then (ii) all the paths which originate from source S and terminate at SW_c . Every path between SW_c and D is found from destination D backward, along a connection at a stage in the reverse direction, one stage after another until stage α is reached.

Since switch D in stage n (the last stage) is connected to three switches in stage $n-1$, namely, switches $D+1 \pmod{N}$, D , and $D-2^{n-1} \pmod{N}$, a path can reach D from any one of these three switches in stage $n-1$, giving rise to three possible paths from stage $n-1$ to D . Similarly, any one of the three switches in stage $n-1$ in turn is connected to three switches in stage $n-2$, resulting in nine possible paths from stage $n-2$ to D . This process repeats, and the number of possible paths to D triples when moving one stage backwards, as a switch j in stage i , $\alpha < i \leq n$, is connected to exactly three switches in the immediate prior stage, stage $i-1$, which are switches $j+1 \pmod{N}$, j , and $j-2^{i-1} \pmod{N}$. In FTGIN_n^α , there are $3^{n-\alpha}$ total paths from stage α to destination D .

Consider a given tag value T . During the process of going backwards from D , the tag value T is changed by a certain amount at each move to reflect the difference $\pmod{N}$ between the current position and S (rather than between D and S). The amount changed depends on the connection taken at that move. Specifically, the amount is 1 (or -2^i) if the connection from switch j in stage $i+1$ to switch $j+1$ (or $j-2^i$) in stage i is taken; it is 0 if the connection from switch j in stage $i+1$ to switch j in stage i is used. Every path from D backward to a switch in stage α , switch SW_c , determines an accumulated amount δ , dictated by the sequence of connections that form the path; this path is referred to as λ . The tag value $T + \delta$ reflects the difference between S and location SW_c in stage α . Since the connecting patterns from stage 0 to stage α are identical to those of the GIN, the number of possible paths from S to SW_c can be estimated using the formula developed for the GIN. Any such possible path may concatenate path λ to form a complete path from S to D , and all the paths so formed are distinct with one another. For a given path λ , there are $\Phi_\alpha(T + \delta)$ total distinct paths between S and D , where Φ_α is defined in Eq. (1) for the GIN and δ is the tag amount changed when path λ is chosen. Let A be the set of those $3^{n-\alpha}$ paths from stage α to D in FTGIN_n^α , then, the total number of alternative

paths for a given tag T in FTGIN_n^α is expressed by

$$\Omega_n^\alpha(T) = \sum_{\lambda \in A} \Phi_\alpha(T + \delta). \quad (2)$$

It is obvious that Eq. (2) becomes Eq. (1) when α equals n , and Eq. (2) gives the same value for $n = 1$ and for $n = 0$. Every path in an FTGIN, like in the GIN, is associated with a tag representation. Since there are totally 3^n different tag representations for any FTGIN_n^α , the summation of alternative paths over all T 's, $0 \leq T < 2^n$ (note that T and $T - 2^n$ are equivalent), should amount to 3^n , regardless of the value of α . This can readily be shown by using Eq. (2). As a result, the FTGIN_n^α distributes those 3^n tag representations to 2^n possible T 's in a way determined by α . Ω_n^α and Ω_n^β for every tag under various α values are provided in Tables I and II. The number of paths between every (S, D) pair in any FTGIN_n^α , as in any FTGIN_n^α , is greater than or equal to 2; multiple paths always exist from any S to any D in an FTGIN. The total number of alternative paths for all tag values in an FTGIN_n^α amounts to 3^n (this fact is confirmed by the results listed in Tables I and II).

Evaluating the Pin Count

The pin counts required for the GIN and the FTGIN are different. Suppose that $r, r > 1$, rows of switches in a network are fabricated in one single chip. There are r switches in stage i of the GIN housed in a chip and those r switches connect to $\eta = 2 \times \min(2^i, r)$ switches of stage $i+1$ which locate outside the chip. The pin count contributed by the connections originating from stage i and terminating at stage $i+1$ is thus equal to η , where min is the minimum function. Likewise, the r switches in stage $i+1$ need to connect η switches of stage i which locate outside the chip, contributing η to the pin count. The number of total pins needed for connections per chip for GIN_n is $2r + 4 \times \sum_{i=0}^{n-1} \min(2^i, r)$, where the first term accounts for the connections for r sources and r destinations.

The pin count for FTGIN_n^α depends not only on n but also on α . For a given n , a smaller α leads to a lower pin count. The amount of total pins for connections per chip for FTGIN_n^α can be derived in a similar way, yielding the pin count of $2r + 2 \times \sum_{i=0}^{n-1} \min(2^i, r) + 2 \times \sum_{i=0}^{\alpha-1} \min(2^i, r) + 2(n - \alpha)$.

A pin count savings always results, and the savings increases as either n or r grows. The savings is non-negligible even for a medium-sized network. When r equals 4, for example, 84 pins per chip are needed for GIN_n , but only 66 pins per chip are required for FTGIN_n^α . This pin savings coupled with shrunk layout area for connections in an FTGIN could translate to cost reduction, making the FTGIN practically attractive.

4. Terminal Reliability Estimation

The number of multiple paths between every (S, D) pair in an FTGIN is not fixed, but rather varying with the

value of $D - S$, i.e., T . It is interesting to examine the terminal reliability of a given T , and compare the result with that of a compatible GIN.

While all the alternative paths for any T in the GIN may be represented by a simple ladder diagram [2], which makes it possible to estimate exact terminal reliability, the alternative paths in an FTGIN no longer constitute a simple structure, because they as a whole may pass through more than two switches in a stage (as opposed to the ladder structure which contains at most two switches in a stage). To derive an exact expression for the terminal reliability of an FTGIN seems fairly complicated. Fortunately, a lower bound on terminal reliability can be attained with less effort. The lower bound, though not very tight, indicates significant improvement over the exact terminal reliability of the GIN in all cases. This large terminal reliability improvement is due mainly to the fact that disjoint paths are guaranteed between every (S, D) pair in an FTGIN.

An *internal switch* refers to a switch located in an internal stage i , $0 < i < n$. Our reliability lower bound is derived by exploring two sets of paths such that the paths in one set and the paths in the other set have no internal switch in common. Consider a *degenerative* FTGIN_n^α obtained from FTGIN_n^α by removing the upper connections between stage i and stage $i+1$, for all $\alpha \leq i < n$, namely, by omitting all connections that are specified by functions $f_i^m(j) = j - 1 \pmod{N}$. Let DGIN_n^α represent the degenerative FTGIN_n^α so obtained, then, DGIN_n^α has the same connecting patterns as GIN_n until stage α , and is without any upper connections from stage α on. Let Ξ denote the set of all possible paths from S to D in DGIN_n^α . It is obvious that any path in Ξ is a path existing from S to D in FTGIN_n^α .

Next, we consider all the paths in DGIN_n^α that originate from S and terminate at switch $D+1 \pmod{N}$ of stage $n-1$. These paths can reach destination D , if a connection from switch $D+1 \pmod{N}$ of stage $n-1$ to switch D in the last stage is present. Let Ψ be the collection of all these paths and ς be such a connection. Assuming that Ψ is the set of paths: $\{\lambda \mid \lambda \equiv \bar{\lambda} \text{ concatenates } \varsigma \& \text{ switch } D \text{ in the last stage, for all } \bar{\lambda} \in \bar{\Psi}\}$. Since a connection ς always exists in FTGIN_n^α , $\alpha < n$, it is apparent that if path λ belongs to Ψ , λ is also present between S and D in FTGIN_n^α . The paths from source 1 to destination 0 of DGIN_n^α which belong to sets Ξ and Ψ are shown in Fig. 4. It is clear that these paths are a portion of all possible paths existing between the $(1, 0)$ pair in FTGIN_n^α given in Fig. 3(b). Furthermore, no path in Ξ shares a common internal switch with any path in Ψ , as described generally in the next theorem.

Theorem: Consider the paths from S to D in an FTGIN_n^α . Any path in Ξ and a path in Ψ have no internal switch in common.

Proof: Let switch j be referred to as an even (odd) switch, if j is even (odd). We observe that in a DGIN (just like in the GIN), an even (odd) switch in stage 0 connects an odd (even) switch in stage 1 when an upper or a lower connection is taken, and connects an even (odd)

switch in stage 1 when a straight connection is taken. In addition, from stage 1 on, an even (odd) switch can reach only even (odd) switches in later stages, regardless of what connections are used, since in a DGIN, switch j in stage i is connected to at most three switches in stage $i+1$, namely, switches $j+2^i \pmod{2^n}$, j , and $j-2^i \pmod{2^n}$, which are all even (odd) switches.

First, let us consider the case when both S and D are even (odd). Because the tag $D-S$ is even, a path in Ξ takes a straight connection, arriving at an even (odd) switch in stage 1. Based on our observation above, the path can reach only even (odd) switches in subsequent stages. On the other hand, a path in Ψ takes either an upper or a lower connection and visits an odd (even) switch in stage 1, since the tag now is $D+1-S$, an odd value. According to the above observation, the path can solely reach odd (even) switches in stages i , $1 < i < n$.

Next, we consider the case when S being even (odd) and D being odd (even). Since the tag $D-S$ is odd, a path in Ξ takes either an upper or a lower connection, arriving at an odd (even) switch in stage 1, and reaches only odd (even) switches in subsequent stages. In contrast, a path in Ψ takes a straight connection and visits an even (odd) switch in stage 1, because the tag now is $D+1-S$, an even value. Apparently, the path can solely reach even (odd) switches in stages i , $1 < i < n$.

Therefore, a path in Ξ and a path in Ψ never have an internal switch in common. The theorem is thus proved. $\square$

If a path belongs to Ξ or Ψ , it must be a path from S to D in an FTGIN $_n^a$ as well. Conversely, a path from S to D in an FTGIN $_n^a$ could belong to neither Ξ nor Ψ . Terminal reliability between the (S, D) pair in an FTGIN $_n^a$ is thereby bounded from below by terminal reliability due to the paths in $\Xi \cup \Psi$.

Terminal reliabilities contributed by the paths in Ξ and by the paths in Ψ are calculated separately. We observe that all the paths in Ξ form a degenerative ladder structure derived from (regular) ladder diagram by removing several links from stage α on, so do the paths in Ψ (recall that the ladder diagram represents all the paths between an (S, D) pair in the GIN). Degenerative ladder structures representing the paths in Ξ between the $(1, 0)$ pair and between the $(1, 4)$ pair are depicted in Fig. 5(b); The degenerative ladder structures representing the paths in Ψ between the same two pairs are simpler and are not given. As before, faults are assumed to happen randomly and independently, and switches in the first and last stages are perfectly reliable.

The terminal reliability of a degenerative ladder structure can be obtained in a way similar to what was described in [2] for the ladder diagram. As an example, consider the degenerative ladder structure representing the multiple paths between the $(1, 0)$ pair shown in Fig. 5(b). Its terminal reliability is expressed by

$$(R(S \rightarrow A) \times p^2 + R(S \rightarrow B) \times (1 - p^2) \times p' \times p^3) \times p',$$

where p and p' are the reliabilities of an internal switch and a switch in the two end stages, respectively, and $R(S \rightarrow A)$ (or $R(S \rightarrow B)$) denotes the terminal reliability between source S and switch A (or switch B). The reliability expression for $R(S \rightarrow A)$ can be derived straightforwardly.

From the nature of GINs, it is readily shown that there are exactly three links respectively with weights $\bar{1}$, 0 , and 1 present between two consecutive stages (after a certain stage dictated by the first nonzero tag bit position) of any ladder structure, as can be found in Fig. 5(a). In fact, the three links existing between any two stages of a ladder structure form only two possible configurations, as illustrated in Fig. 5(c). This results from the basic properties of routing in GINs: (i) if a link with weight $\bar{1}$ from any switch is on a path, then the link with weight $\bar{1}$ from the same switch is also on a possible path, and (ii) at most two switches in each stage are involved in all the paths between any communication pair. This regularity in the ladder structure makes it possible to systematically determine a general degenerative ladder structure. The general degenerative ladder structure after stage α is given in Fig. 5(d), where the dotted links existing in the regular ladder structure are absent in the degenerative ladder structure. A degenerative ladder structure has two parallel path segments from stage α on, until a stage where the two path segments get merged. In general, if the three links in the regular ladder structure are in the form as shown in the left of Fig. 5(c), the absence of links with weight $\bar{1}$ in the corresponding degenerative ladder structure yields two parallel path segments. On the other hand, if the three links are in the form as shown in the right of Fig. 5(c), then the two parallel path segments merge afterwards, giving rise to a single path segment from there on.

Any degenerative ladder structure is obtained from its corresponding regular ladder structure, if the merged point after stage α is decided. Let β be greater than α and be the first zero bit position in the tag $(= D - S \pmod{N})$ for the (S, D) pair. Then, the merged point lies in stage $\beta+1$, as depicted in Fig. 5(d), and two parallel link segments exist between stages α and $\beta+1$. The terminal reliability of the obtained degenerative ladder structure, which denotes all the paths in Ξ , can be calculated. Similarly, the degenerative ladder structure representing all the paths in Ψ can be constructed and its terminal reliability is evaluated. Let terminal reliabilities due to the paths in Ξ and due to the paths in Ψ be denoted respectively by R_Ξ and R_Ψ . Terminal reliability due to all the paths in $\Xi \cup \Psi$ is equal to $R_\Xi + R_\Psi - R_\Xi \times R_\Psi$, as switches in the first and last stages are assumed fault-free.

The terminal reliability lower bound so calculated versus tag for various FTGINs with the system sizes of 16 and 64 is plotted in Fig. 6, where the reliability of a 3×3 switch is assumed to be 0.9 as before. Compared with the GIN, an FTGIN enjoys a significant terminal reliability enhancement, particularly at those tags where the GIN suffers from relatively low reliability values. When the tag is 32 $(= N/2)$ in a size 64 network, for instance, terminal reliability is raised from roughly 0.59 to more than 0.87 for

FTGIN⁸, clearly indicating the advantage of creating disjoint paths between vulnerable communication pairs. It is observed from Fig. 6(a) and Fig. 6(b) that the terminal reliability gap between GIN_n and FTGIN_{n-v} grows as n increases, for any tag value and $v < n$, signifying that a larger FTGIN leads to more improvement.

5. Conclusions

In this paper, we have investigated "modified" Gamma interconnection networks obtained simply by altering connecting patterns between certain consecutive stages of the Gamma network. A "modified" network ensures the existence of totally disjoint paths between every source-destination pair and is dubbed as an FTGIN. Routing complexity is the same for an FTGIN as for its GIN counterpart. All the paths from any source to any destination in an FTGIN can be represented by a "modified" binary redundant number system. Results on the distribution of paths for different tags in FTGINs are derived. If several rows of switching elements are fabricated in one chip using VLSI technology, an FTGIN could lead to reduced cost, because the pin count per chip decreases and the layout area taken by connections shrinks. An FTGIN is shown to exhibit significant terminal reliability improvement over the conventional Gamma network. We believe that due to its substantially enhanced reliability and cost reduction potential, an FTGIN could be a good candidate for interconnecting medium to large scale high-performance multiprocessors.

References

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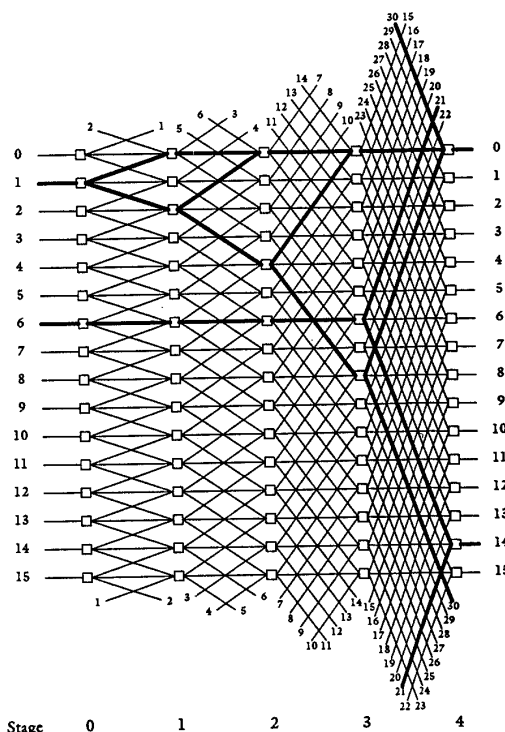


Fig. 1. A Gamma interconnection network with size 16.
(Bold lines indicate alternative paths between (1, 0) and between (6, 14).)

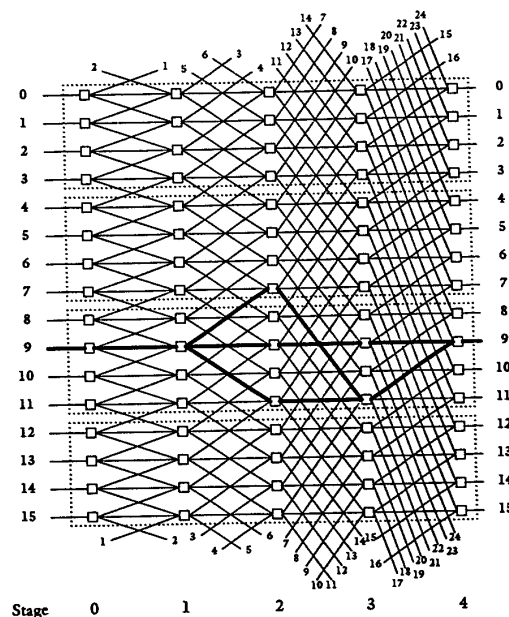
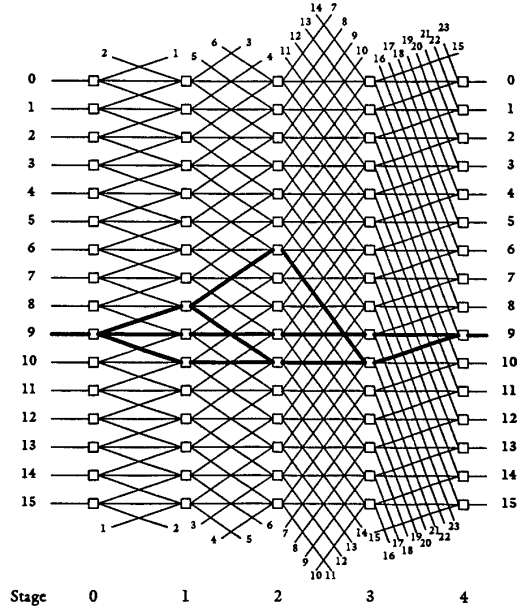
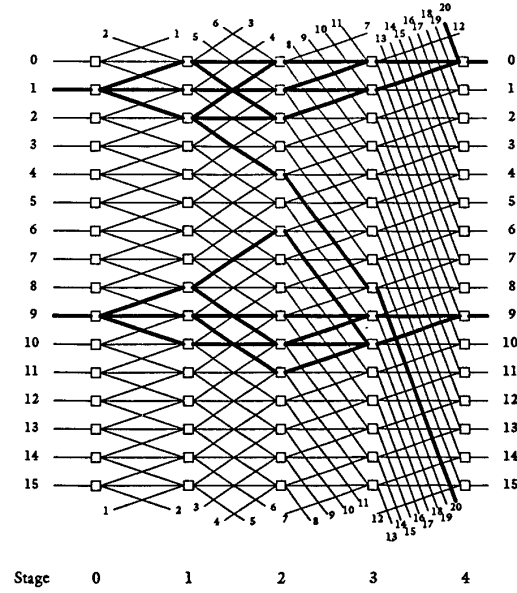


Fig. 2. A "modified" Gamma interconnection network.
(Switches within a dotted rectangle are fabricated in one chip.)



(a) FTGIN_4^3



(b) FTGIN_4^2

Fig. 3. Two possible fault-tolerant GINs under consideration. (Alternative paths between the (9, 9) pair are highlighted in both networks, and the paths between the (1, 0) pair in FTGIN_4^2 are also shown.)

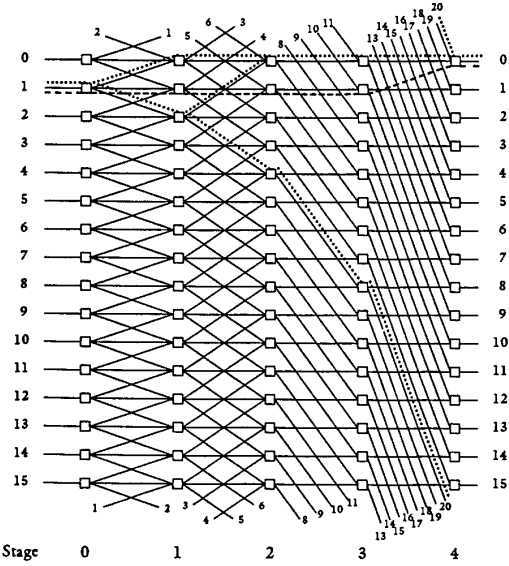


Fig. 4. The degenerative network of FTGIN_4^2 given in Fig. 3(b). (All paths existing between the (1, 0) pair are highlighted in a way that the paths in Ξ and in Ψ are denoted respectively by dotted and dashed lines.)

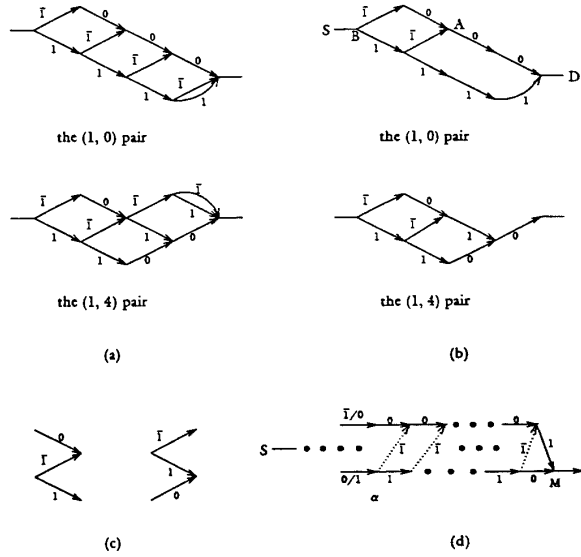


Fig. 5. (a) Ladder diagrams representing all paths between two given communication pairs in GIN_4 . (b) Degenerative ladder structures representing all paths between the given two pairs in DGIN_4^2 shown in Fig. 4. (c) Two possible path configurations at any stage. (d) The general degenerative ladder structure illustrating the merged point M after stage α .

Table I. The number of alternative paths for every tag in a network with size 16

Tag	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
GIN_4	1	5	4	7	3	8	5	7	2	7	5	8	3	7	4	5
FTGIN_4^3	4	6	6	6	4	6	4	4	2	5	5	6	5	7	6	5
FTGIN_4^2	7	7	6	4	4	5	5	4	3	4	3	4	4	7	7	7

(The number of total paths amounts to $3^4=81$ in every network.)

Table II. The number of alternative paths for every tag in a network with size 64

Tag	GIN_6	$FTGIN_6^5$	$FTGIN_6^4$	$FTGIN_6^3$	$FTGIN_6^2$
0	1	6	13	22	27
1	7	10	17	24	23
2	6	12	18	22	18
3	11	12	17	20	17
4	5	12	19	20	19
5	14	16	20	18	19
6	9	14	17	13	16
7	13	12	13	10	14
8	4	10	14	12	14
9	15	16	17	15	13
10	11	16	17	15	11
11	18	16	14	14	11
12	7	12	13	15	14
13	17	16	13	15	16
14	10	12	10	12	15
15	13	10	6	9	14
16	3	6	6	10	14
17	14	12	9	12	13
18	11	12	11	12	10
19	19	14	10	10	8
20	8	10	11	10	9
21	21	16	13	10	10
22	13	12	12	8	9
23	18	12	9	6	7
24	5	6	9	7	7
25	17	12	12	10	7
26	12	10	13	11	7
27	19	12	11	11	8
28	7	6	10	12	11
29	16	10	11	13	14
30	9	6	9	11	14
31	11	6	6	9	14
32	2	2	5	10	14
33	11	7	8	12	13
34	9	7	9	12	10
35	16	10	9	10	8
36	7	7	8	10	9
37	19	13	11	10	10
38	12	10	9	8	9
39	17	11	8	5	7
40	5	7	5	5	7
41	18	14	9	7	7
42	13	13	8	8	6
43	21	15	9	7	5
44	8	10	5	7	6
45	19	15	8	8	8
46	11	11	5	7	8
47	14	10	6	5	7
48	3	7	4	4	7
49	13	13	9	6	7
50	10	14	9	6	6
51	17	15	12	6	4
52	7	13	11	4	4
53	18	18	15	6	5
54	11	15	12	5	5
55	15	13	13	8	4
56	4	10	13	9	3
57	13	15	18	14	5
58	9	15	17	15	8
59	14	14	17	19	15
60	5	11	16	20	20
61	11	13	17	21	25
62	6	10	13	18	26
63	7	7	11	19	28

(The number of total paths amounts to $3^6 = 729$ in every network.)

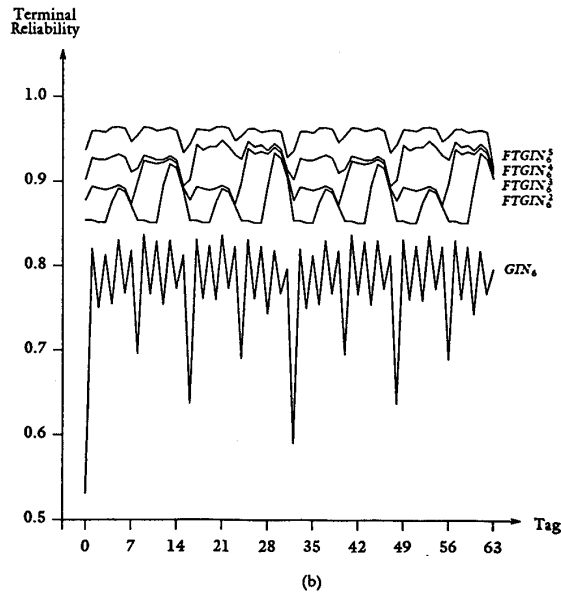
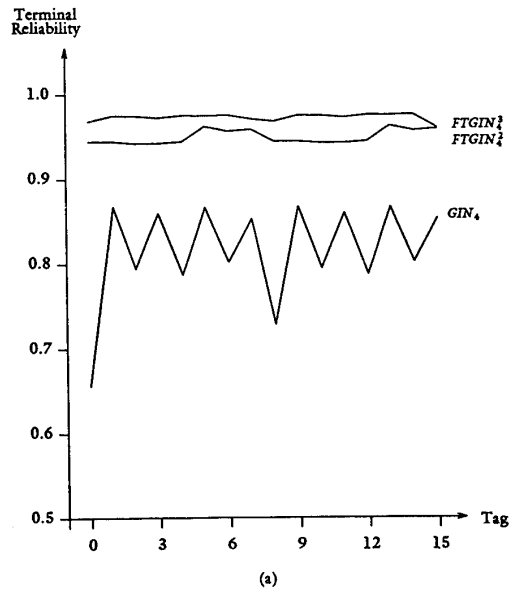


Fig. 6. Terminal reliability comparison among (a) GIN_4 and various $FTGIN_4^p$'s, and (b) GIN_6 and various $FTGIN_6^p$'s.

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